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Coaxial cable attenuation in NMR sample coil circuits

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An analysis is presented of the effect on power transfer and noise figure of attenuation in the radiofrequency coaxial cable (coax) between a NMR sample coil and the associated spectrometer. A matching network between the sample coil and the coax, which matches the coil to the coax characteristic impedance, is shown to decrease the noise temperature of the coax by a factor of the Q of the coil. In addition, the usual two-capacitor matching network, which provides a perfect match only if both capacitors are variable, is shown to yield an acceptably small mismatch over a wide range of frequencies even if one capacitor is fixed.

INTRODUCTION

A pulsed nuclear magnetic resonance (NMR) experiment involves transmission of high-power radiofrequency (rf) pulses from a power amplifier (the "transmitter") to a coil wrapped around the sample, followed by detection of the resulting nuclear resonance signal by a low-noise preamplifier and signal processor (the "receiver").¹ If the NMR experiment is carried out at a remote location (in a low-temperature cryostat, a furnace, a superconducting magnet, etc.), these rf signals travel through a coaxial transmission line (coax) between the sample coil and the NMR spectrometer.

The rf attenuation by the coax is small but not negligible: At 50 MHz attenuations of $0.2\text{--}0.3\text{ dB m}^{-1}$ for semirigid stainless-steel coax and $0.08\text{--}0.14\text{ dB m}^{-1}$ for semirigid copper coax are common.² Part of the design procedure for a NMR system is to minimize the effects of attenuation in the coax, subject to other considerations such as minimization of receiver recovery time, cost, complexity, etc. The specific effects to be minimized are (1) loss of rf magnetic field amplitude B_1 in the sample coil for a given transmitter power, and (2) reduction of NMR signal-to-noise (S/N) ratio at the receiver input due to additional noise generated in the coax. The latter is often the more important consideration. The loss will be greater if stainless-steel coax must be used, as in a cryostat where the coax must have low thermal conductance.³

Optimization of these properties is usually carried out by matching the NMR coil to both the transmitter and the receiver electronics. This is done by transforming the sample coil impedance $Z_s = R_s + i\omega L_s$ to a standard impedance (often the $50\text{-}\Omega$ characteristic impedance of many coaxial cables) by means of low-loss reactive elements. This form of matching makes use of the resonant properties of LC circuits and is often referred to as "tuning" the coil. Specific matching schemes have been shown to maximize both B_1 and the S/N ratio^{1,4} if the same coil is used for both transmission and reception; this "single-coil" geometry^{4,5} is commonly used because it affords the best filling factor and has no serious disadvantages compared to a two-coil configuration.

Two matching schemes are in wide use. In the "resonant-coax" scheme, shown in Fig. 1(a), the NMR sample

coil is connected directly to the coax, and matching is accomplished at the other end.^{1,3,5} The coax thus forms part of the resonant circuit: It is not itself matched, and a standing wave is present on it. This scheme is simple and is often used if the S/N ratio is high, transmitter power is ample, and little room is available for additional circuit components near the sample.

The "matched-load" scheme, shown in Fig. 1(b), uses capacitors in a matching network between the sample coil

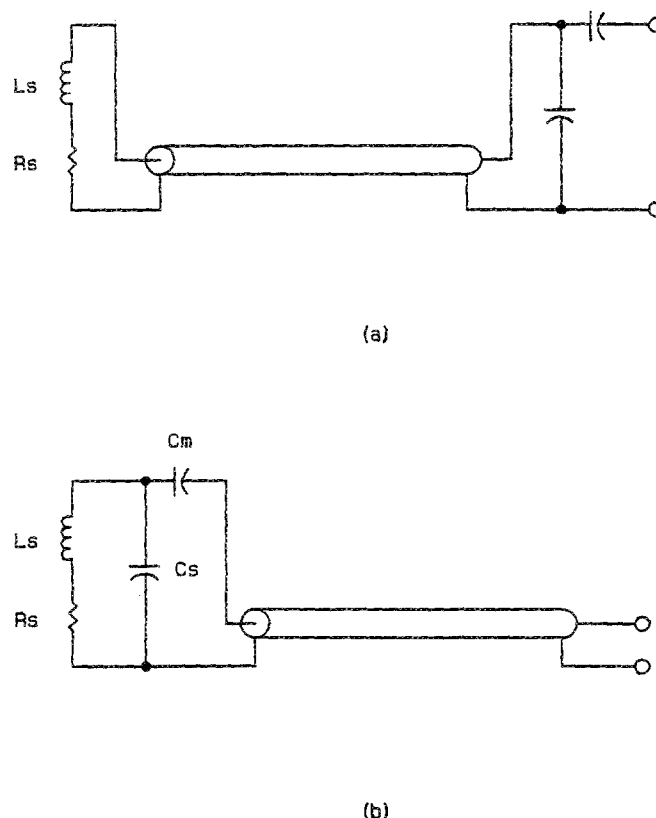


FIG. 1. Matching of NMR sample coil (inductance L_s , series resistance R_s) through a coaxial cable. (a) "Resonant-coax" scheme. A capacitor divider matching network is shown, but other networks can be used (see, e.g., Ref. 5). (b) "Matched-load" scheme, using a capacitor divider (sample-coil tuning capacitance C_s and matching capacitance C_m). The matching networks in (a) and (b) are assumed lossless.

and the coax.^{4,6-10} The coax is then not part of the resonant circuit, and effects of coax attenuation are considerably reduced at the cost of placing a matching network near the sample. The improved performance available from the matched-load scheme has been known qualitatively for some time,⁶⁻¹⁰ but a general analysis has not to the author's knowledge been previously reported.

The purpose of this article is to consider in some detail optimization of the rf properties of a NMR system in which the coax attenuation is significant. Section I describes power dissipation in the system, the noise temperature T_{nc} associated with the coax and the effective quality factor Q_{eff} . Section II treats general power transfer to a load which may or may not be matched to the coax. The specific matching schemes described above are analyzed in Sec. III. In Sec. IV it is shown that use of the matched-load scheme of Fig. 1(b) with one of the matching capacitors fixed (which simplifies the matching network at the sample coil) yields only a moderate reduction of performance over a wide range of frequencies. This is related to the system of Walton and Conradi,¹¹ in which neither of the matching capacitors is variable, and is of interest when NMR experiments must be carried out over a range of frequencies and space near the sample is limited. Section V describes experimental and numerical studies of the results of this analysis and discusses some of the trade-offs involved in practical system design.

I. POWER LOSS, NOISE FIGURE, AND QUALITY FACTOR

Figure 2 gives schematic diagrams of the network we wish to analyze in its two modes of operation: the "transmit" mode [Fig. 2(a)] and the "receive" mode [Fig. 2(b)]. (So-called "duplexing" methods for switching between these modes have been described in the literature^{1,3,5} and will not be discussed here.) The sample coil, of inductance L_s , resistance R_s , and quality factor $Q_s = \omega L_s / R_s$, is connected to a two-port network, which represents the coax and any reactive matching elements.

In the transmit mode [Fig. 2(a)] the transmitter, represented by a Thévenin equivalent (complex) generator voltage V_g and series impedance Z_g , provides an available mean power $P = |V_g|^2 / 4Z_g$. A current I_1 flows in the sample coil, causing a fraction $g_s P$ of the transmitter power to be dissipated in the sample coil resistance. The current produces a linearly polarized rf magnetic field $2B_1$ in the coil, the magnitude of which is given in SI units by

$$2B_1 / I_1 = [2\mu_0 g_s P Q_s / (V\omega)]^{1/2}. \quad (1)$$

Here $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ is the permeability of free space, and V is an effective sample volume including the effects of the filling factor and fringing fields.^{4,5,12}

In the receive mode [Fig. 2(b)] the signal voltage V_s induced in the sample coil by the precessing nuclear magnetization results in an available mean signal power $P_s = |V_s|^2 / 4R_s$. This signal power is coupled to the input impedance Z_{in} of the receiver through the coax and matching network. In addition, Johnson noise voltage V_n in the coil resistance R_s gives rise to an available (and inevitable) noise power $N_s = BkT_s$, where B is the system bandwidth and T_s

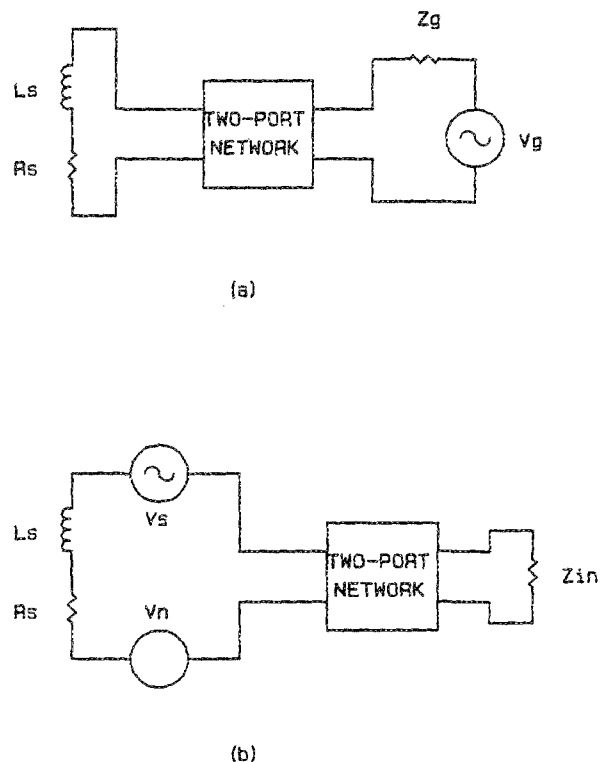


FIG. 2. Schematic diagram of NMR sample coil network. (a) "Transmit" mode: Power from the transmitter voltage source V_g is coupled to the sample coil resistance R_s . (b) "Receive" mode: Power from the signal and noise voltage sources V_s and V_n is coupled to the input impedance Z_{in} of the receiver. The two-port network represents the coax and any matching elements.

is the physical temperature of the coil. For a given coil geometry the intrinsic (power) S/N ratio P_s / N_s of the coil can be increased only by increasing its Q or by decreasing its temperature.⁴ Thus, maximization of the S/N ratio for fixed coil Q and temperature is equivalent to minimizing the additional noise at the receiver input due to Johnson noise from the dissipative elements of the coax. Noise contributed by the NMR spectrometer electronics must also be minimized; apart from a few general remarks we will not consider this noise source further.

Analysis of this circuit is most easily carried out using "Pierce's rule,"¹³ which relates the noise properties of a network to the distribution of power dissipation in the network. Suppose that power input P at a port of a linear passive network results in a power P_i dissipated in block i of the network. Equivalently, suppose that a fraction $g_i = P_i / P$ of the input power is dissipated in block i , with $\sum g_i = 1$. We will call g_i the power gain from the input to block i ($g_i < 1$), and the power loss l_i is defined by $l_i = 1/g_i$. Then Pierce's rule states that the contribution of Johnson noise from block i to the noise power available at the port is $N_i = g_i BkT_i$ where T_i is the physical temperature of block i .

In Fig. 2 we divide the network into two blocks: (1) the sample coil, and (2) the two-port network, containing the coax and any reactive matching elements. The latter are taken to be lossless. Input power is then dissipated in (1) the sample coil resistance (gain g_s), and (2) the coax (gain g_c

$= 1 - g_s$). Referring to Fig. 2(b), the NMR signal and the noise generated in the coil resistance can both be represented by voltage sources (V_s and V_n , respectively) in series with the coil. Now Pierce's rule states that the available noise power at the receiver is reduced by the factor g_s . The available signal power, which is equivalent in its circuit characteristics to the available noise power, must be reduced by the same factor.

The coax attenuation does not *per se* change the S/N ratio. Additional noise power will, however, be contributed by the coax if it is lossy and its temperature is finite. Following standard treatments¹³ the contribution N_c of the coax to the available noise power at the receiver input is represented by an equivalent noise temperature T_{nc} referred to the sample coil. By definition an increase T_{nc} of the physical temperature of the coil would result in an increase N_c of output noise power, or $N_c = g_s BkT_{nc}$. By Pierce's rule $N_c = g_c BkT_c = (1 - g_s) BkT_c$, where T_c is the physical temperature of the coax, so that T_{nc} is given by

$$T_{nc}/T_c = \frac{1 - g_s}{g_s} = l_s - 1, \quad (2)$$

where l_s is the sample-coil power loss. From the usual definition¹³ the noise factor of the system is l_s and the noise figure in dB is $20 \log_{10} l_s$. The available output noise power from the entire system is then $N_{out} = g_s BkT_{out}$, where

$$T_{out} = T_s (1 + T_{nc}/T_s), \quad (3)$$

so that the ratio T_{nc}/T_s is a measure of the increase in noise due to the coax.

Thus, it suffices to calculate g_s (or l_s) for the circuit of Fig. 2(a) to obtain (1) the loss of transmitter power and (2) the decrease of NMR signal, both due to coax attenuation, as well as (3) the Johnson noise power added by the coax. These properties can be determined independently of whether or not the load is matched to the coax.

It is clear from the foregoing that full advantage of the high S/N ratios inherent in NMR experiments at low temperatures is achieved only if T_{nc} and the noise temperature T_{nr} of the spectrometer receiver^{4,6} are smaller than T_s . On the other hand, the NMR signal follows Curie's law and is large at low temperatures, so that noise minimization may be less important. The noise temperature of the entire system is the sum of T_{nc} and T_{nr} ; once $T_{nc} \ll T_{nr}$ further improvements in system performance require reduction of the latter. Commercially available low-noise wide-band rf amplifiers¹⁴ typically have noise figures between 1.5 and 2.5 dB or, equivalently, noise temperatures between 120 and 220 K, whereas homemade preamplifiers cooled in liquid nitrogen can achieve noise figures as low as 0.2–0.3 dB (15–20 K).^{6,15} The trade-off between noise figure and system complexity must be decided separately for each specific experiment.

The peak energy E_s stored in the magnetic field of the sample coil is proportional to B_1^2 , and is given by $E_s = \frac{1}{2} L_s |I_1|^2$. By definition the Q of the sample coil is the ratio of E_s to the mean energy E_d dissipated in R_s in 1 rad of angular frequency ω . Then, $E_d = P_s / \omega$, and $Q_s = \omega E_s / P_s$. An effective quality factor Q_{eff} for the network as a whole is then

$$Q_{eff} = \omega E_s / P, \quad (4)$$

where electromagnetic energy storage in the coax is not included because we wish to use Q_{eff} to characterize power transfer to the coil. If the coax is lossless $Q_{eff} = Q_s$, and in general

$$Q_{eff} = Q_s (P_s / P) = g_s Q_s. \quad (5)$$

Optimization of power transfer is therefore equivalent to maximization of Q_{eff} .

II. POWER TRANSFER BY LOSSY COAX

We begin by recalling several aspects of standard transmission-line theory.¹⁶ If the terminating load impedance does not have a particular resistive value, called the characteristic impedance Z_c , a portion of an incident wave traveling toward the load will be reflected. Let V_0^+ and I_0^+ be the (complex) incident voltage and current, respectively, at the load, and let V_0^- and I_0^- be the reflected voltage and current. The reflection coefficient ρ_0 at the load is then defined by

$$\rho_0 = V_0^- / V_0^+ = I_0^- / I_0^+. \quad (6)$$

The voltage and current ratios are equal because for the incident and reflected waves separately the ratio of voltage to current is just Z_c .¹⁶ It is convenient to express ρ_0 in polar form: $\rho_0 = |\rho_0| e^{i\psi_0}$.

Then at any point a distance x from the load the voltage V_x and current I_x will be given by

$$V_x = V_x^+ + V_x^- = V_0^+ e^{\gamma x} + V_0^- e^{-\gamma x}, \quad (7)$$

and

$$I_x = I_x^+ - I_x^- = I_0^+ e^{\gamma x} - I_0^- e^{-\gamma x}. \quad (8)$$

Here, the propagation constant $\gamma = \alpha + i\beta$, where α is the attenuation per unit length and β is the phase shift per unit length. The minus signs in Eq. (8) are due to the 180° phase difference between voltage and current phase changes at reflection. For $x = 0$, Eqs. (7) and (8) give the impedance Z_0 of the load:

$$Z_0 = V_0 / I_0 = Z_c \left(\frac{1 + \rho_0}{1 - \rho_0} \right), \quad (9)$$

or, equivalently,

$$\rho_0 = \frac{Z_0 - Z_c}{Z_0 + Z_c}. \quad (10)$$

From Eqs. (6) and (7) the reflection coefficient $\rho_x = V_x^- / V_x^+$ at point x is

$$\rho_x = \rho_0 e^{-2\gamma x}, \quad (11)$$

so that

$$V_x = V_x^+ (1 + \rho_x) = V_0^+ (e^{\gamma x} + \rho_0 e^{-\gamma x}) \quad (12)$$

and

$$I_x = I_x^+ (1 - \rho_x) = I_0^+ (e^{\gamma x} - \rho_0 e^{-\gamma x}). \quad (13)$$

We now consider the ratio of the mean power P_x supplied by a generator to a coax of length x to the mean power P_0 dissipated in a load, of complex impedance Z_0 , at the other end of the coax. This ratio is by definition the load power loss, and it will also be the sample-coil power loss l_s if

the matching network in the matched-load scheme [Fig. 1(b)] is made of lossless reactances.

We use the fact that a voltage V across and a current I into a complex impedance dissipates a mean power $\frac{1}{2} \Re(V^* I)$, where $\Re$ indicates the real part. Therefore,

$$\begin{aligned} P_x &= \frac{1}{2} \Re(V_x^* I_x) \\ &= P_x^+ \Re[(1 + \rho_x^*)(1 - \rho_x)] \\ &= P_x^+ (1 - |\rho_x|^2), \end{aligned} \quad (14)$$

with $P_x^+ = \frac{1}{2} \Re[(V_x^+)^* I_x^+]$. In particular,

$$P_0 = P_0^+ (1 - |\rho_0|^2). \quad (15)$$

From Eqs. (7), (8), and (11), $P_x^+ = P_0^+ e^{2\alpha x}$ and $|\rho_x| = |\rho_0| e^{-2\alpha x}$, so that

$$\begin{aligned} P_x/P_0 &= I_s \\ &= \frac{e^{2\alpha x} - |\rho_0|^2 e^{-2\alpha x}}{1 - |\rho_0|^2}. \end{aligned} \quad (16)$$

Note that this result is independent of the coax phase factor βx , i.e., of the electrical length of the coax. If $\alpha x \ll 1$, as will be the case for any reasonable attenuation factor, Eq. (16) becomes

$$I_s = 1 + \left(\frac{1 + |\rho_0|^2}{1 - |\rho_0|^2} \right) 2\alpha x. \quad (17)$$

Using Eq. (10) we can write this in terms of the load impedance $Z_0 = R_0 + iX_0$:

$$I_s = 1 + \left(\frac{R_0^2 + X_0^2 + Z_c^2}{R_0 Z_c} \right) \alpha x. \quad (18)$$

We shall make extensive use of Eq. (18) in the following.

III. APPLICATION TO SPECIFIC MATCHING SCHEMES

A. Resonant-coax scheme

In this technique the sample coil is not matched to the coax, and the load impedance is $Z_0 = Z_s = R_s + i\omega L_s$.

We first consider the case where the coil resistance is negligible: $R_s \rightarrow 0$. Then from Eq. (18) it is clear that I_s diverges, which is reasonable since there is no coil resistance in which to dissipate power. We consider instead $Q_{\text{eff}}^{-1} = I_s/Q_s$ [Eq. (5)], which from Eq. (18) is

$$Q_{\text{eff}}^{-1} = Q_s^{-1} + \left(\frac{X_s^2(1 + Q_s^{-2}) + Z_c^2}{X_s Z_c} \right) \alpha x. \quad (19)$$

Then, Q_{eff}^{-1} is minimized with respect to variations in X_s when $X_s^2 = Z_c^2/(1 + Q_s^{-2})$, for which

$$\begin{aligned} Q_{\text{eff}}^{-1} &= Q_s^{-1} + (1 + Q_s^{-2})^{1/2} 2\alpha x \\ &\approx Q_s^{-1} + 2\alpha x, \end{aligned} \quad (20)$$

or, equivalently,

$$\begin{aligned} I_s &= 1 + (1 + Q_s^2)^{1/2} 2\alpha x \\ &\approx 1 + 2Q_s \alpha x. \end{aligned} \quad (21)$$

The approximate relations in Eqs. (20) and (21) hold for $R_s \ll X_s$, $Q_s \gg 1$, for which the minimization condition is $Z_c = X_s$ as has been noted previously.³ Minimization with re-

spect to X_s for fixed Q_s is appropriate, because sample coils with different numbers of turns but otherwise similar geometries have similar Q 's.⁴

B. Matched-load scheme

From Eq. (18) if the coil is perfectly matched to the coax $R_0 = Z_c$ and $X_0 = 0$, so that $\rho_0 = 0$. Then,

$$I_s = 1 + 2\alpha x, \quad (22)$$

and

$$Q_{\text{eff}}^{-1} = Q_s^{-1} (1 + 2\alpha x). \quad (23)$$

For large Q_s the effect of the coax attenuation is reduced by a factor Q_s^{-1} compared to the resonant-coax scheme [Eqs. (20) and (21)]. It is this reduction that has been previously noted qualitatively.⁶⁻¹⁰

Figure 1(b) shows a typical capacitor-divider matching network. The operation of this network is easily understood.⁴ The "tuning" capacitance C_s is adjusted so that the resonant frequency $\omega_s = 1/\sqrt{L_s C_s}$ of the parallel combination ($C_s || L_s$) is higher than the operating frequency ω , and of a value such that the resistive component of ($C_s || L_s$) is equal to the coax characteristic impedance. Then, the "matching" capacitance C_m is adjusted to cancel the residual inductive reactance of ($C_s || L_s$).

IV. MATCHED-LOAD SCHEME WITH FIXED MATCHING CAPACITOR

If sample space is at a premium it may be inconvenient to place two variable capacitors near the sample coil. If a fixed matching capacitor C_m were used, however, the matching would be perfect ($\rho_0 = 0$) for only one frequency ω_0 , but might be approximately correct ($|\rho_0| \ll 1$) for a range of nearby frequencies. This suggests that a modest mismatch will not reduce system performance by a large factor, at least compared to the difference between the matched-load and resonant-coax schemes. It is worthwhile considering this situation in some detail, given the potential simplification of a variable-frequency NMR system.

We approximate the sample coil by a resistanceless inductance L_s in parallel with a resistance $R_p = Q_s^2 R_s$ which accounts for dissipation in the coil. Near resonance ($\omega_s \approx \omega$) and for $Q_s \gg 1$, the impedance of the network which terminates the coax is approximately

$$\begin{aligned} Z_0 &= R_0 + iX_0 \\ &= \frac{R_p}{1 + 2iQ_s \delta} + iX_m \\ &= R_p \left(\frac{1 - iy}{1 + y^2} \right) + iX_m, \end{aligned} \quad (24)$$

where $\delta = (\omega - \omega_s)/\omega$ and $y = 2Q_s \delta$. The conditions for perfect matching are then

$$R_0 = \frac{R_p}{1 + y^2} = Z_c, \quad (25)$$

and

$$X_0 = X_{m0} - \frac{yR_p}{1+y^2} = 0. \quad (26)$$

Now consider the situation where X_m differs by a small amount ΔX_m from the perfectly matched value X_{m0} given by Eq. (26). A new value of y can then be found such that the system is optimized: In practice, this would be done by varying C_s to minimize the coax loss. To find the minimization condition we use Eq. (24) for R_0 and X_0 in Eq. (18), and obtain

$$l_s(y) = 1 + \left(\frac{R_p^2 - 2yR_pX_m + (X_m^2 + Z_c^2)(1+y^2)}{R_pZ_c} \right) \alpha x. \quad (27)$$

Then at the minimum

$$y = y_{\min} = \frac{X_m R_p}{X_m^2 + Z_c^2}, \quad (28)$$

for which

$$(l_s)_{\min} = \left(\frac{X_m^2 + Z_c^2}{R_p Z_c} + \frac{R_p Z_c}{X_m^2 + Z_c^2} \right) \alpha x. \quad (29)$$

Using Eqs. (25), (26), and (29), it is straightforward to confirm our previous result $(l_s)_{\min} = 1 + 2\alpha x$ for $X_m = X_{m0}$. With $X_m = X_{m0} + \Delta X_m$, and for $R_p \gg Z_c$ (high sample-coil Q), we have to first nonvanishing order in ΔX_m

$$(l_s)_{\min} = 1 + \left[1 + 2 \left(\frac{\Delta X_m}{X_{m0}} \right)^2 \right] 2\alpha x. \quad (30)$$

This shows that degradation of the system performance by incorrect values of X_m (i.e., $\omega \neq \omega_0$ for fixed C_m) is moderate as long as the tuning capacitance C_s is readjusted to minimize power loss.

V. EXPERIMENTAL AND NUMERICAL RESULTS

This article has treated the performance of a rf network for NMR experiments, in which attenuation in the coaxial cable between the sample coil and the NMR spectrometer is an important feature of the problem. It is concluded that use of a matching network at the sample-coil end of the coax (the matched-load scheme), compared to matching only at the spectrometer end of the coax (the resonant-coax scheme), reduces the effect of coax attenuation by a factor of the Q of the sample coil. Furthermore, the simplest such network provides an acceptable match over a wide range of frequencies with only one tuning adjustment.

Some experimental and numerical examples are useful to illustrate the performance of these matching schemes. The ^{63}Cu nuclear quadrupole resonance (NQR) in Cu_2O , which has a frequency of about 26 MHz at room temperature,¹⁷ was used for this purpose. The sample coil had an inductance of $\sim 1.2 \mu\text{H}$ and a Q of between 15 and 25, and the sample-coil circuit included a 1.5-m length of a standard 0.141-in. (3.58-mm) stainless-steel coax (Uniform Tubes² type UG-141SS). The attenuation factor αx for this cable is specified to be approximately 0.04 at 26 MHz. NQR free induction decays were measured using the matched-load and resonant-coax schemes. From the amplitudes of the NMR signals the ratio of the power loss factors for the two schemes was determined to be 4.0 ± 0.2 . The noise temperature of the system, $\sim 150 \text{ K}$, was dominated by spectrometer receiver noise, and the experimental noise temperature of the coax could not be determined separately. Even in such a case the S/N ratio is improved, often considerably, by using the matched-load scheme to reduce the loss of signal power in the coax.

Table I gives calculated values of the sample-coil power loss l_s , the coax noise figure, and coax noise temperatures for physical coax temperatures of 300 and 77 K, for a system with the parameters of our experimental apparatus and $Q_s = 20$. The calculated ratio of resonant-coax to matched-load power loss is somewhat smaller than the observed value. However, the inductance of the coil used in our experiment was rather far from the value of $0.3 \mu\text{H}$ required for the optimum value of coil reactance ($\omega L_s = Z_c$ at 26 MHz) in the resonant-coax scheme. The calculated resonant-coax power loss when the actual inductance is used in Eq. (21) is ~ 6 for $Q_s = 20$, which is in reasonable agreement with the observed value given the uncertainty in the sample-coil Q . The large difference in noise temperature between the resonant-coax and matched-load schemes is evident. In a cryostat the majority of the coax length will typically be at 300 K unless precautions are taken to lower the coax temperature by thermal shunting to a liquid-nitrogen bath, in which case the coax noise temperature will be correspondingly reduced.

We next consider the matched-load scheme with only the tuning capacitance C_s [Fig. 1(b)] adjustable. Typically, the available frequency range from minimum to maximum value of C_s is about a factor of 2. From Eq. (30) a $\pm 50\%$ error in the matching reactance X_m (i.e., a $\pm 50\%$ variation of frequency around the value ω_0 which yields perfect matching) increases the coax noise temperature by only $\sim 50\%$. Figure 3 gives the calculated frequency dependence

TABLE I. Comparison of matched-load and resonant-coax matching schemes for typical system parameters. Sample-coil $Q_s = 20$; sample-coil inductance $L_s = 0.3 \mu\text{H}$; frequency for perfect match $\omega_0/2\pi = 26 \text{ MHz}$; coax attenuation factor $\alpha x = 0.043$.

	Coax temp. T_c (K)	Matched load	Resonant coax
Power loss l_s	...	1.09	2.73
Noise figure (dB)	...	0.4	4.4
Coax noise temp. T_{nc} (K) ^b	300	25.9	519
	77	6.6	133

^aDependent on coax temperature only through temperature dependence of α .

^bAssuming α independent of coax temperature.

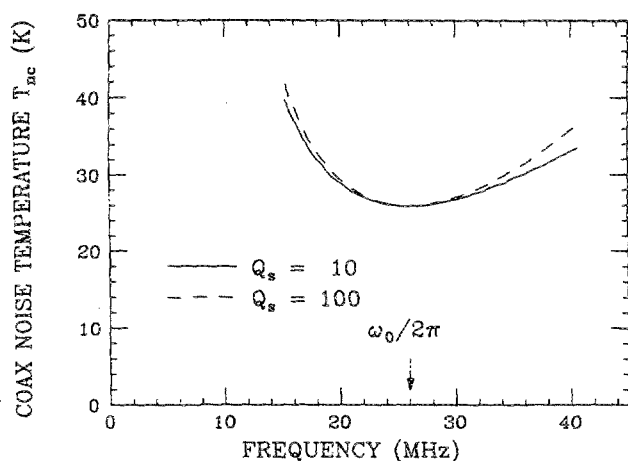


FIG. 3. Frequency dependence of coax noise temperature T_{nc} using the matched-load scheme with a fixed matching capacitor, for coax attenuation factor $\alpha x = 0.043$ and coax physical temperature $T_c = 300$ K. Solid curve: sample-coil quality factor $Q_s = 10$. Dashed curve: $Q_s = 100$. The frequency $\omega_0/2\pi$ for a perfect match is 26 MHz.

of the noise temperature T_{nc} of our system for different values of Q_s , obtained from Eqs. (2) and (29). It can be seen that the increase of T_{nc} above the minimum value is in accord with the above rough estimate and depends only slightly on Q_s . Experimental measurements of power gain with fixed C_m confirmed these results qualitatively by showing less than a few percent change over a 2:1 range of frequency variation. As noted above the experimental system noise figure was too large to allow a direct measurement of the coax noise temperature.

If experiments are to be carried out over a wide range of temperatures Q_s will vary, typically by a factor of 3–4 between 300 and 77 K and somewhat more slowly at lower temperatures if the coil is wound from unannealed copper wire. An increase of Q_s turns out to decrease the matching capacitance C_m for perfect matching at a given frequency ω_0 or, equivalently, to decrease the value of ω_0 for perfect matching at fixed C_m . This could prove bothersome if the signal is weak and the best S/N ratio is required. One recourse would be to change C_m for experiments in different temperature ranges; another would be to choose a compromise value of C_m such that ω_0 sweeps through the operating frequency as the temperature is lowered. Mallory polystyrene SX-series and Centralab type 854/855 capacitors have been reported⁹ to be satisfactory for use at cryogenic temperatures.

One possible modification to improve the performance of a resonant-coax system in a low-temperature cryostat would be to install copper coax over the majority of the distance between the coil and room temperature, with a length of stainless-steel coax at the top for thermal isolation. A relatively short length (~ 50 cm) of UG-141SS coax results in ~ 40 mW heat flow from 300 to 4 K. This would be appreciable but probably acceptable; it would result in an additional liquid-helium boiloff of ~ 50 cm³ h⁻¹. The copper coax would, of course, be cold over its entire length, but the heat leak to the cryogen could be minimized by use of a small-

diameter outer conductor (e.g., Uniform Tubes UG-85) and careful attention to insulation.

Alternatively, a home-made coax cable could be used in which the inner surface of the stainless-steel outer conductor is copper plated to a somewhat greater thickness than the rf skin depth. This would restore the lower attenuation factor of copper compared to stainless steel. But in commercial semirigid coax² the PTFE insulation is bonded to the conductors and provides a good vacuum seal at the top of the cryostat. This is not available in the homemade cable, and a separate seal must be constructed. In addition, glow discharge can occur in the helium bath vapor during the transmitter pulse if the high-voltage rf lead is exposed; care must be taken to avoid this.

The performance of either of these arrangements would be improved at low temperatures, by reduction of the coax attenuation factor (by about a factor of ~ 3 below 77 K) and, in the case of all-copper coax, reduction of the coax physical temperature. But compared to the matched-load scheme the system Q would deteriorate near or above room temperature.

Typically, the S/N ratio of the spectrometer is the most important consideration; efficient use of transmitter power is usually secondary. It is worthwhile to note that if signal averaging is used the time required to average to a given final S/N ratio is directly proportional to the overall noise temperature of the spectrometer (including sample-coil noise), assuming all other factors are constant. Once the noise temperature of the cryostat rf network is reduced to well below that of the receiver electronics, further reduction is of little use.

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